

Math Thermo
class # 19
3/31/2026

Conservation Equations

Definition (19.1): Suppose that

$$\rho: \Omega \rightarrow [0, \infty)$$

is the mass density of a non-uniform substance. Assume that G is an extensive variable of the substance in Ω such that, for every $V \subseteq \Omega$,

$$G_V = \int_V \rho(\vec{x}) g(\vec{x}) d\vec{x},$$

where

$$g: \Omega \rightarrow \mathbb{R}$$

is called the specific density relative to G . This framework is called the specific density framework.

Example (19.2): Suppose that S is the entropy and U is the internal energy of a non-uniform substance in Ω . Then

$$S_V = \int_V \rho(\vec{x}) s(\vec{x}) d\vec{x}$$

$$U_V = \int_V \rho(\vec{x}) u(\vec{x}) d\vec{x}$$

for any $V \subseteq \Omega$, where $s: \Omega \rightarrow \mathbb{R}$ is the specific entropy density (entropy per unit mass) and $u: \Omega \rightarrow \mathbb{R}$ is the specific internal energy density (internal energy per unit mass).

$$[s] = \frac{\text{entropy}}{\text{mass}} = \frac{\text{Joules}}{\text{deg. Kelvin} \cdot \text{Kilogram}}$$

$$[u] = \frac{\text{Joules}}{\text{Kilogram}}$$

$$[\rho s] = \frac{\text{entropy}}{\text{volume}} = \frac{\text{Joules}}{\text{deg. Kelvin} \cdot \text{m}^3}$$

$$[\rho u] = \frac{\text{Joules}}{\text{m}^3}$$

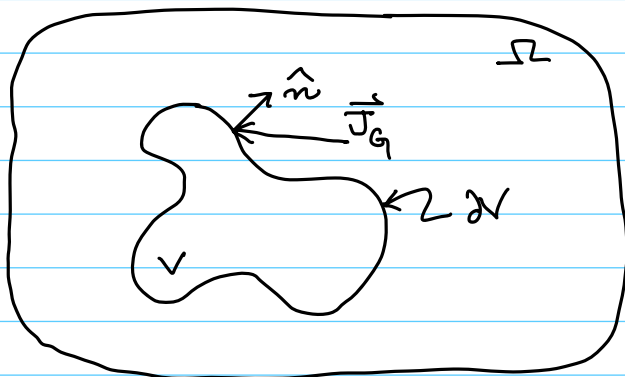
To find a conservation equation for G , assume the fields depend upon time and take the time derivative

$$\frac{d}{dt} G_v(t) = \int_V \frac{\partial}{\partial t} (\rho g) d\vec{x}$$

$$= - \int_{\partial V} \vec{J}_G \cdot \hat{n} dS + \int_V \sigma_G d\vec{x}$$

$$= \int_V \{ -\nabla \cdot \vec{J}_G + \sigma_G \} d\vec{x},$$

using the divergence theorem. The picture is



$$\vec{J}_G = \text{flux of } \rho g$$

$$\sigma_G = \text{source of } \rho g \text{ in } \Omega$$

Since $V \subseteq \Omega$ is arbitrary

(19.1)

$$\partial_t(\rho g) = -\nabla \cdot \vec{J}_g + \sigma_g.$$

Suppose that G is the total mass of our material. In this case, we take

$$G = M, \quad g = 1.$$

Then

$$M_v = \int_V \rho(\vec{x}) d\vec{x}.$$

We assume that total mass is neither created nor destroyed pointwise. Thus,

$$\sigma_M = 0.$$

Furthermore, we assume that

$$\vec{J}_M = \rho \vec{v},$$

where $\vec{v}$ is the mass-averaged velocity, which we will define momentarily. Then

(19.2)

$$\partial_t \rho + \nabla \cdot (\rho \vec{v}) = 0.$$

This fundamental equation is called the barycentric mass continuity equation.

We can derive this equation for a mixture.

Suppose that there are $r \in \mathbb{N}$ chemicals in a mixture. Let

$$\rho_i : \Omega \rightarrow [0, \infty), \quad i = 1, \dots, r,$$

be the mass densities of each component. Each component satisfies a mass conservation equation of the form

$$(19.3) \quad \partial_t \rho_i + \nabla \cdot (\rho_i \vec{v}_i) = \sigma_i.$$

Where $\vec{v}_i$ is the convective velocity of species i . Define

$$\vec{v} := \frac{\sum_{i=1}^r \rho_i \vec{v}_i}{\rho}$$

where

$$\rho(\vec{x}, t) := \sum_{i=1}^r \rho_i(\vec{x}, t)$$

is the total mass density. This $\vec{v}$ is called the mass-averaged or barycentric velocity.

We assume that the mass sources sum to zero:

$$\sum_{i=1}^r \sigma_i = 0.$$

Thus, if we sum (19.3), we get

$$\partial_t \rho + \nabla \cdot (\rho \vec{v}) = \sum_{i=1}^r \sigma_i = 0,$$

which is the continuity equation (19.2).

We will often use another form for (19.3). let us define

$$-\vec{J}_i^D := \rho_i(\vec{v} - \vec{v}_i).$$

Then,

$$\begin{aligned} (19.4) \quad \partial_t \rho_i + \nabla \cdot (\rho_i \vec{v}) &= \nabla \cdot (\rho_i(\vec{v} - \vec{v}_i)) + \sigma_i \\ &= -\nabla \cdot \vec{J}_i^D + \sigma_i \end{aligned}$$

$\vec{J}_i^D$ is called the diffusion flux of chemical i .
Clearly,

$$\begin{aligned} -\sum_{i=1}^r \vec{J}_i^D &= \sum_{i=1}^r \rho_i(\vec{v} - \vec{v}_i) \\ &= \rho \vec{v} - \sum_{i=1}^r \rho_i \vec{v}_i \\ &= \rho \vec{v} - \rho \vec{v} \\ &= \vec{0}. \end{aligned}$$

Thus, only $r-1$ diffusion fluxes can be independent:
for example

$$\vec{J}_1^D = -\sum_{i=2}^r \vec{J}_i^D.$$

Definition (19.3): Suppose that in an r -component fluid ρ is the total mass and $\vec{v}$ is the mass-averaged velocity,

$$\rho \vec{v} = \sum_{i=1}^r \rho_i \vec{v}_i.$$

The material derivative relative to the mass-averaged velocity is defined as

$$(19.5) \quad \frac{D_\vec{v} f}{Dt} = \frac{Df}{Dt} := \frac{\partial f}{\partial t} + \vec{v} \cdot \nabla_\vec{x} f$$

for any function

$$f: \Omega \times [0, T] \rightarrow \mathbb{R}.$$

let us go back to our general conservation equation (19.1), but let us write it in a different, equivalent form:

$$(19.6) \quad \frac{\partial(\rho g)}{\partial t} + \nabla \cdot (\rho g \vec{v}) = -\nabla \cdot \vec{J}_g^D + \sigma_g$$

where we have written

$$\vec{J}_g = \underbrace{\rho g \vec{v}}_{\substack{\uparrow \\ \text{barocentric convective flux}}} + \underbrace{\vec{J}_g^D}_{\substack{\uparrow \\ \text{diffusive flux}}}$$

Note that there is nothing magical about

the pairing ρg . We could equally well use

$$G_V = \int_V \tilde{g}(\vec{x}) d\vec{x}$$

to express the extensive quantity G over the volume $V \subseteq \Omega$. Then, we would have

$$\frac{\partial \tilde{g}}{\partial t} + \nabla \cdot (\tilde{g} \vec{v}) = -\nabla \cdot \vec{J}_G^D + \sigma_G.$$

Going back to (19.6),

$$\begin{aligned} \frac{\partial(\rho g)}{\partial t} &= \rho \frac{\partial g}{\partial t} + g \frac{\partial \rho}{\partial t} \\ &\stackrel{(19.2)}{=} \rho \frac{\partial g}{\partial t} - g \nabla \cdot (\rho \vec{v}) \\ &= \rho \frac{\partial g}{\partial t} - \nabla \cdot (\rho g \vec{v}) + \rho \vec{v} \cdot \nabla g \end{aligned}$$

Thus,

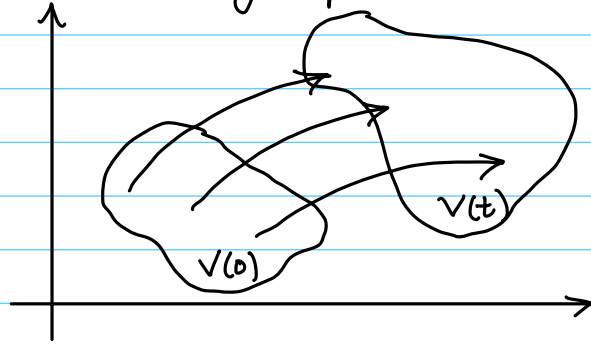
$$(19.7) \quad \rho \frac{Dg}{Dt} = \frac{\partial(\rho g)}{\partial t} + \nabla \cdot (\rho g \vec{v})$$

Therefore, the general conservation law for the extensive quantity G can be written as

$$(19.8) \quad \boxed{\rho \frac{Dg}{Dt} = -\nabla \cdot \vec{J}_G^D + \sigma_G.}$$

The identity (19.7) is, essentially, the integral form of a well known result that we now state:

Theorem (Reynolds Transport Theorem): Suppose that $V(t) \subseteq \Omega$ is a subset of Ω that moves according to velocity field $\vec{w}$



Then,

$$\frac{d}{dt} \int_{V(t)} \tilde{q} d\vec{x} = \int_{V(t)} \left\{ \frac{\partial \tilde{q}}{\partial t} + \nabla \cdot (\tilde{q} \vec{w}) \right\} d\vec{x}$$

If $\vec{w}$ is the barycentric velocity, $\vec{v}$, then

$$\begin{aligned} \frac{d}{dt} \int_{V(t)} \rho g d\vec{x} &= \int_{V(t)} \left\{ \frac{\partial (\rho g)}{\partial t} + \nabla \cdot (\rho g \vec{v}) \right\} d\vec{x} \\ &= \int_{V(t)} \frac{D(\rho g)}{Dt} d\vec{x}. \end{aligned}$$

Proof: See, for example,

Gonzalez and Stuart "A First Course in Continuum Mechanics," Cambridge (2008).